

EE390: Digital Systems Engineering

Homework 3

1. Write an 8051 C program to toggle all bits of P1 every 200 ms. (Assume the single iteration of the inner loop typically takes about **12 machine cycles**)
 2. Write an 8051 C program to count up P1 from 0 - 99 continuously.
 3. Find the contents of the port after each of the following operations.
 - i. $P1 = 0x65 \& 0x76;$
 - ii. $P1 = 0x70 | 0x6B;$
 - iii. $P2 = 0x95 \wedge 0xAA;$
 - iv. $P2 = 0x5D \& 0x78;$
 - v. $P2 = 0xC5 | 0x12;$
 - vi. $P0 = 0x6A \wedge 0x6E;$
 - vii. $P1 = 0x37 | 0x26;$
 - viii. $P1 = 0x65 \gg 2;$
 - ix. $P2 = 0x39 \ll 2;$
 - x. $P1 = 0xD4 \gg 3;$
 - xi. $P1 = 0xA7 \ll 2;$
 4. Assume that XTAL = 16 MHz. Find the TH1 and TL1 values required to generate a time delay of 5 ms. Timer 1 is programmed in Mode 1.
 5. Assuming that XTAL = 11.0592 MHz and a square wave is generated on pin P1.2, find the lowest square-wave frequency that can be generated using Mode 1.
 6. Program Timer 0 to generate a square wave of 3 kHz. Assume that XTAL = 11.0592 MHz.
 7. Program Timer 1 in C to generate a square wave of 3 kHz. Assume that XTAL = 11.0592 MHz.
 8. A switch is connected to pin P1.2. Write an 8051 C program to monitor the switch and generate the following frequencies on pin P1.7:
 - SW = 0: 500 Hz
 - SW = 1: 750 Hz
- Use Timer 0 in Mode 1 for both frequencies.
9. Assume that a 1 Hz pulse signal is connected to input pin P3.4. Write a program to display the Timer 0 count on an LCD. Set the initial value of TH0 to -60.
 10. Assuming that clock pulses are applied to pin T1, write a program to configure Timer 1 as a counter in Mode 2, count the incoming pulses, and display the content of TL1 on Port 2.